Building Diabetes Prediction Models

```
diabetes = readRDS("clean_diabetes_data.rds")
```

Model 1 Preparation

```
df_multilog <- diabetes |>
  select(diabetes_012, high_bp, high_chol, chol_check, bmi, smoker, stroke,
         heart_diseaseor_attack, phys_activity, fruits, veggies, hvy_alcohol_consump,
         any_healthcare, no_docbc_cost, gen_hlth, ment_hlth, phys_hlth, diff_walk,
         sex, age, education, income)
df_multilog
```

```
## # A tibble: 253,680 x 22
##    diabetes_012 high_bp high_chol chol_check   bmi smoker stroke
##    <fct>        <fct>   <fct>     <fct>      <dbl> <fct>  <fct>
##  1 No Diabetes  Yes     Yes       Yes           40 Yes    No
##  2 No Diabetes  No      No        No            25 Yes    No
##  3 No Diabetes  Yes     Yes       Yes           28 No     No
##  4 No Diabetes  Yes     No        Yes           27 No     No
##  5 No Diabetes  Yes     Yes       Yes           24 No     No
##  6 No Diabetes  Yes     Yes       Yes           25 Yes    No
##  7 No Diabetes  Yes     No        Yes           30 Yes    No
##  8 No Diabetes  Yes     Yes       Yes           25 Yes    No
##  9 Diabetes     Yes     Yes       Yes           30 Yes    No
## 10 No Diabetes  No      No        Yes           24 No     No
## # i 253,670 more rows
## # i 15 more variables: heart_diseaseor_attack <fct>, phys_activity <fct>,
## #   fruits <fct>, veggies <fct>, hvy_alcohol_consump <fct>,
## #   any_healthcare <fct>, no_docbc_cost <fct>, gen_hlth <ord>, ment_hlth <dbl>,
## #   phys_hlth <dbl>, diff_walk <fct>, sex <fct>, age <ord>, education <ord>,
## #   income <ord>
```

train/test split

```
# using createDataPartition because diabetes_012 is unbalanced based on EDA.
train_test_split <- createDataPartition(df_multilog$diabetes_012, p = 0.8, list = FALSE)
train <- df_multilog[train_test_split, ]
test  <- df_multilog[-train_test_split, ]

y_test <- test$diabetes_012
```

F1 function for multiclass

```
multi_f1 <- function(true, pred) {
  # Convert to factors
```

```
  true <- as.factor(true)
  pred <- as.factor(pred)

  # make sure all classes are taken into consideration (bc diabetes=1 is rare)
  class <- union(levels(true), levels(pred))
  true <- factor(true, levels = class)
  pred <- factor(pred, levels = class)

  # A vector to store F1 for each class
  multi_f1_vec <- numeric(length(class))

  # Loop over each class
  for (j in seq_along(class)) {
    i <- class[j]
    tp <- sum(true == i & pred == i)
    fn <- sum(true == i & pred != i)
    fp <- sum(true != i & pred == i)
    rec  <- if((tp + fn) == 0) 0 else tp / (tp + fn)
    prec <- if((tp + fp) == 0) 0 else tp / (tp + fp)

    if(prec + rec == 0) {
      multi_f1_vec[j] <- 0
    } else {
      multi_f1_vec[j] <- 2 * prec * rec / (prec + rec)
    }
  }
    mean(multi_f1_vec)
}
```

Model 1: Multinomial Logistic Regression

```
model1 <- multinom(diabetes_012 ~ ., data = train, trace = FALSE)
```

```
model1_pred <- predict(model1, newdata = test, type = "class")
model1_prob <- predict(model1, newdata = test, type = "prob")

model1_accuracy <- mean(model1_pred == y_test)
cat("Accuracy is:", model1_accuracy, "\n")
```

```
## Accuracy is: 0.8481719
```

```
model1_macro_f1  <- multi_f1(y_test, model1_pred)
cat("Macro F1 score is:", model1_macro_f1, "\n")
```

```
## Macro F1 score is: 0.39549
```

```
model1__macro_auc <- as.numeric(multiclass.roc(response = y_test, predictor = model1_prob)$auc)
cat("Macro AUC is:", model1__macro_auc, "\n")
```

```
## Macro AUC is: 0.6955571
```

1.1 ROC

```
cols <- c("blue", "green", "pink")
```

```r
# Compute one-vs-rest ROC and AUC for each class
model1_roc1 <- roc(as.numeric(y_test == levels(y_test)[1]), model1_prob[, levels(y_test)[1]])
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```r
model1_roc2 <- roc(as.numeric(y_test == levels(y_test)[2]), model1_prob[, levels(y_test)[2]])
```

```
## Setting levels: control = 0, case = 1
## Setting direction: controls < cases
```

```r
model1_roc3 <- roc(as.numeric(y_test == levels(y_test)[3]), model1_prob[, levels(y_test)[3]])
```

```
## Setting levels: control = 0, case = 1
## Setting direction: controls < cases
```

```r
# Plot ROC
plot(model1_roc1, col = cols[1], main = "ROC - Multinomial Logistic Regression")
lines(model1_roc2, col = cols[2])
lines(model1_roc3, col = cols[3])

# Compute class AUCs
model1_class_aucs <- sapply(levels(y_test), function(cl) round(auc(roc(as.numeric(y_test == cl), model1
```

```
## Setting levels: control = 0, case = 1
## Setting direction: controls < cases
```

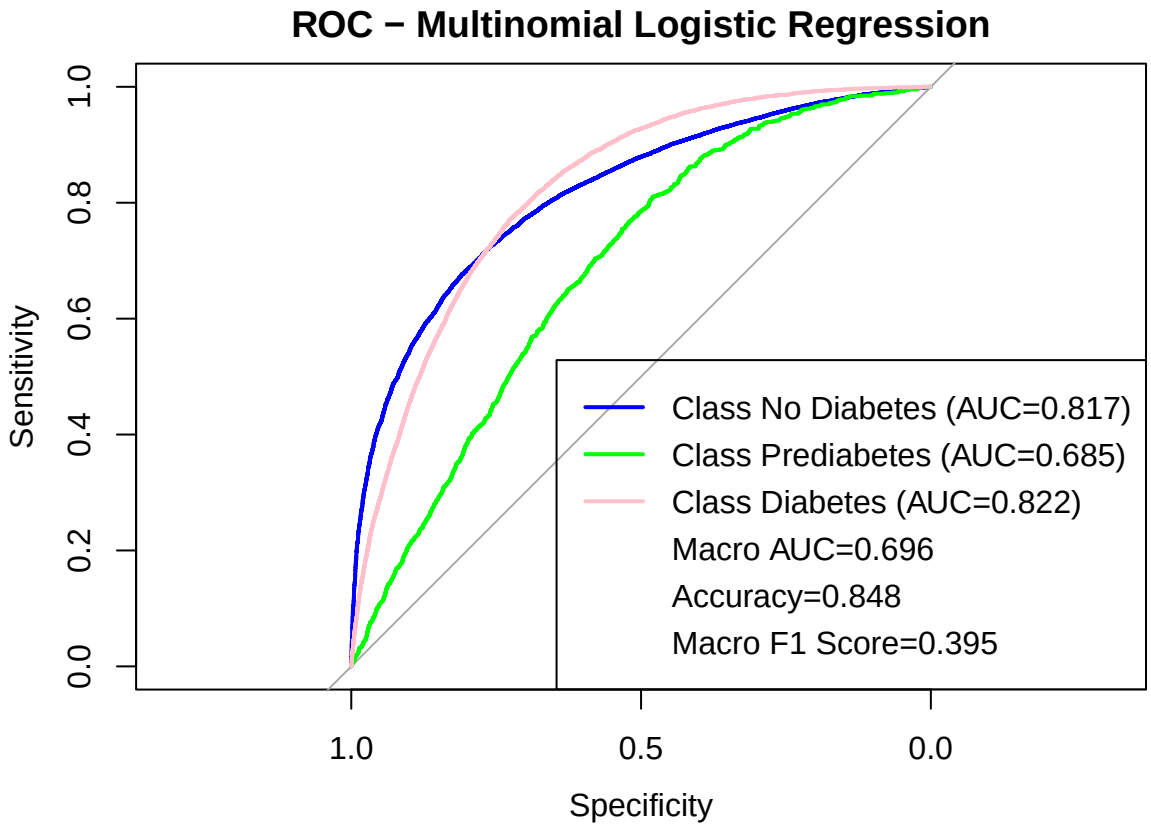
```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```
## Setting levels: control = 0, case = 1
```

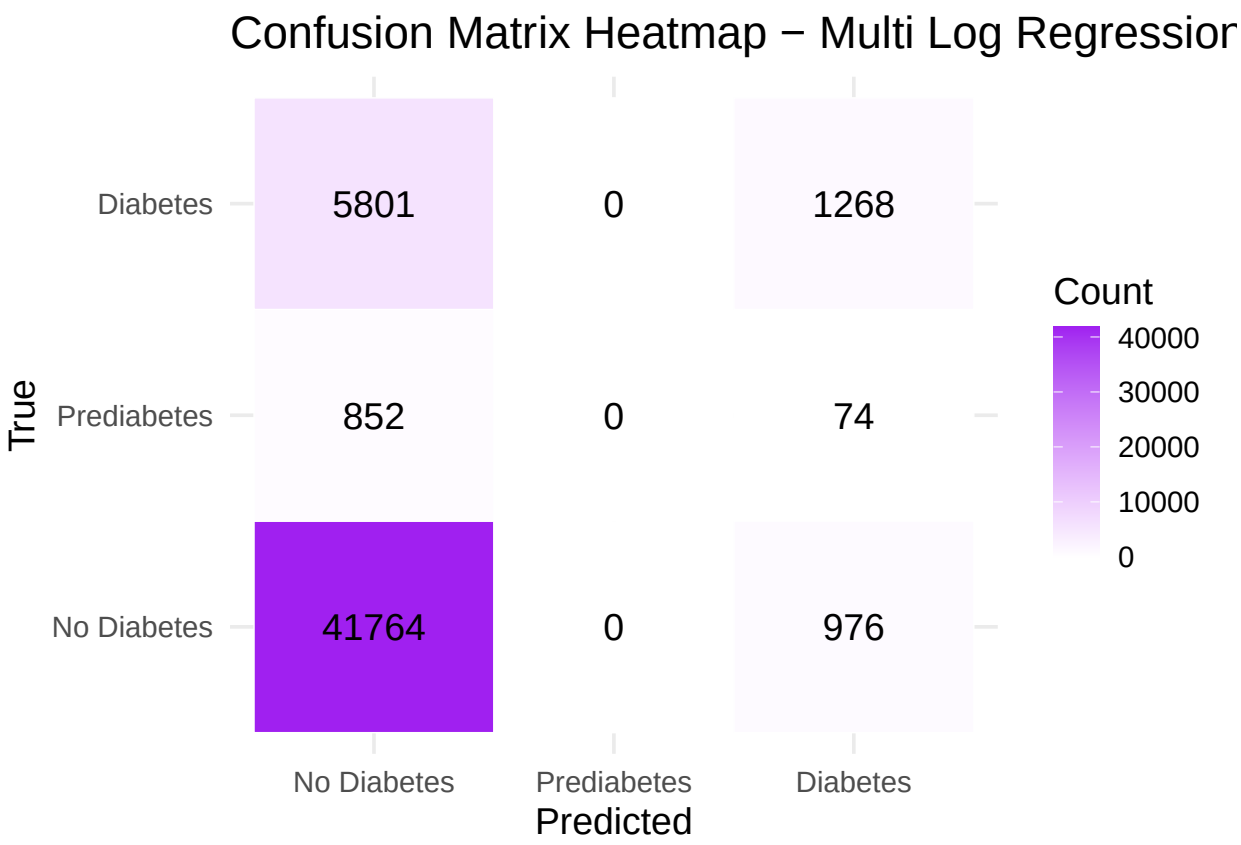
```
## Setting direction: controls < cases
```

```r
# Add legend with class AUCs + macro AUC
legend("bottomright",
       legend = c(paste0("Class ", levels(y_test), " (AUC=", model1_class_aucs, ")"),
                  paste0("Macro AUC=", round(model1__macro_auc, 3)),
                  paste0("Accuracy=", round(model1_accuracy, 3)),
                  paste0("Macro F1 Score=", round(model1_macro_f1, 3))),
       col = c(cols, rep(NA, 3)),
       lwd = c(rep(2, 3), rep(NA, 3)))
```



1.2 Confusion matrix heatmap

```
model1_cm <- table(True = y_test, Pred = model1_pred)
model1_cm_df <- as.data.frame(model1_cm) |>
  rename(True = True, Predicted = Pred, Count = Freq)

ggplot(model1_cm_df, aes(x = Predicted, y = True, fill = Count)) +
  geom_tile(color = "white") +
  geom_text(aes(label = Count), color = "black", size = 5) +
  scale_fill_gradient(low = "white", high = "purple") +
  labs(title = "Confusion Matrix Heatmap - Multi Log Regression", x = "Predicted", y = "True") +
  theme_minimal(base_size = 14)
```



Model 2 Preparation

```
df_ranger <- diabetes |>
  select(diabetes_2, high_bp, high_chol, chol_check, bmi, smoker, stroke,
         heart_diseaseor_attack, phys_activity, fruits, veggies, hvy_alcohol_consump,
         any_healthcare, no_docbc_cost, gen_hlth, ment_hlth, phys_hlth, diff_walk,
         sex, age, education, income)

df_ranger$diabetes_2 <- factor(df_ranger$diabetes_2,
                               levels = c(0, 1),
                               labels = c("No_Diabetes", "Diabetes"))
df_ranger
```

```
## # A tibble: 253,680 x 22
##    diabetes_2  high_bp high_chol chol_check   bmi smoker stroke
##    <fct>       <fct>   <fct>     <fct>      <dbl> <fct>  <fct>
##  1 No_Diabetes Yes     Yes       Yes           40 Yes    No
##  2 No_Diabetes No      No        No            25 Yes    No
##  3 No_Diabetes Yes     Yes       Yes           28 No     No
##  4 No_Diabetes Yes     No        Yes           27 No     No
##  5 No_Diabetes Yes     Yes       Yes           24 No     No
##  6 No_Diabetes Yes     Yes       Yes           25 Yes    No
##  7 No_Diabetes Yes     No        Yes           30 Yes    No
##  8 No_Diabetes Yes     Yes       Yes           25 Yes    No
##  9 Diabetes    Yes     Yes       Yes           30 Yes    No
## 10 No_Diabetes No      No        Yes           24 No     No
## # i 253,670 more rows
```

```
## # i 15 more variables: heart_diseaseor_attack <fct>, phys_activity <fct>,
## #   fruits <fct>, veggies <fct>, hvy_alcohol_consump <fct>,
## #   any_healthcare <fct>, no_docbc_cost <fct>, gen_hlth <ord>, ment_hlth <dbl>,
## #   phys_hlth <dbl>, diff_walk <fct>, sex <fct>, age <ord>, education <ord>,
## #   income <ord>
```

Model 2: Binary Ranger/RandomForest Model

2.1 Data splitting

```
set.seed(1)

index <- createDataPartition(df_ranger$diabetes_2, p = 0.8, list = FALSE)
train_data <- df_ranger[index, ]
test_data <- df_ranger[-index, ]

y_test <- test_data$diabetes_2
```

2.2 Train Binary Random Forest Model (using ranger)

```
rf_model <- ranger(
  dependent.variable.name = "diabetes_2",
  data = train_data,
  num.trees = 500,
  importance = "impurity",
  seed = 1,
  verbose = TRUE,
  probability = TRUE
)
```

```
## Growing trees.. Progress: 65%. Estimated remaining time: 16 seconds.
```

2.3 Prediction and Metrics

```
pred <- predict(rf_model, data = test_data, type = "response")

# Probability of Diabetes (positive class)
model_prob <- pred$predictions[, "Diabetes"]

# Class predictions = class with highest probability
model_pred <- colnames(pred$predictions)[ max.col(pred$predictions) ]
model_pred <- factor(model_pred, levels = levels(y_test))

# Confusion matrix
cm <- confusionMatrix(model_pred, y_test)

# Accuracy
model_accuracy <- as.numeric(cm$overall['Accuracy'])

# F1 for Diabetes
f1_diabetes <- as.numeric(cm$byClass['F1'])
# F1 for No_Diabetes
cm_rev <- confusionMatrix(model_pred, y_test, positive = "No_Diabetes")
f1_no_diabetes <- as.numeric(cm_rev$byClass['F1'])
```

```r
# Macro F1
model_macro_f1 <- mean(c(f1_diabetes, f1_no_diabetes))

# AUC for Diabetes
model_roc <- roc(response = y_test, predictor = model_prob)
```

```
## Setting levels: control = No_Diabetes, case = Diabetes
```

```
## Setting direction: controls < cases
```

```r
model_auc <- as.numeric(model_roc$auc)

cat("Accuracy is:     ", round(model_accuracy, 4), "\n")
```

```
## Accuracy is:      0.8523
```

```r
cat("Macro F1 score is:", round(model_macro_f1, 4), "\n")
```

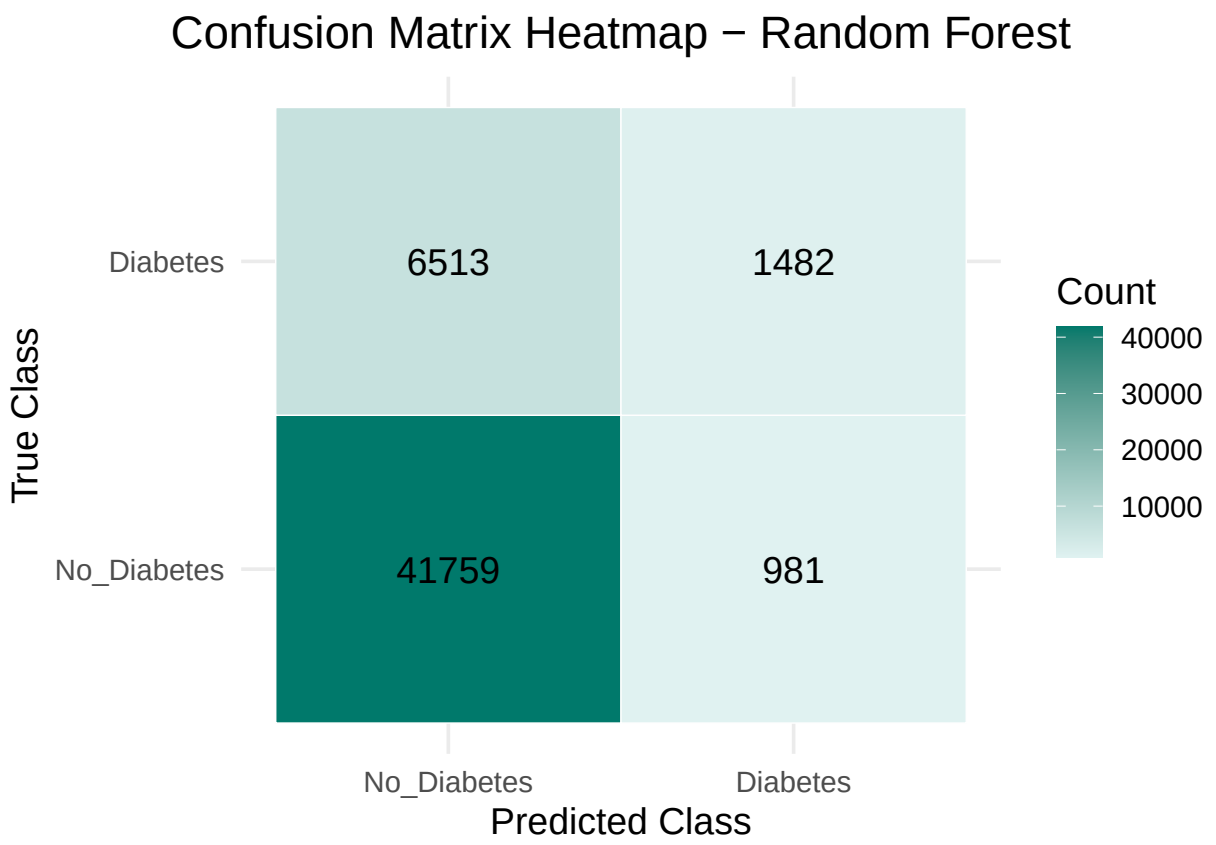
```
## Macro F1 score is: 0.9177
```

```r
cat("ROC AUC (Binary) is:", round(model_auc, 4), "\n")
```

```
## ROC AUC (Binary) is: 0.8195
```

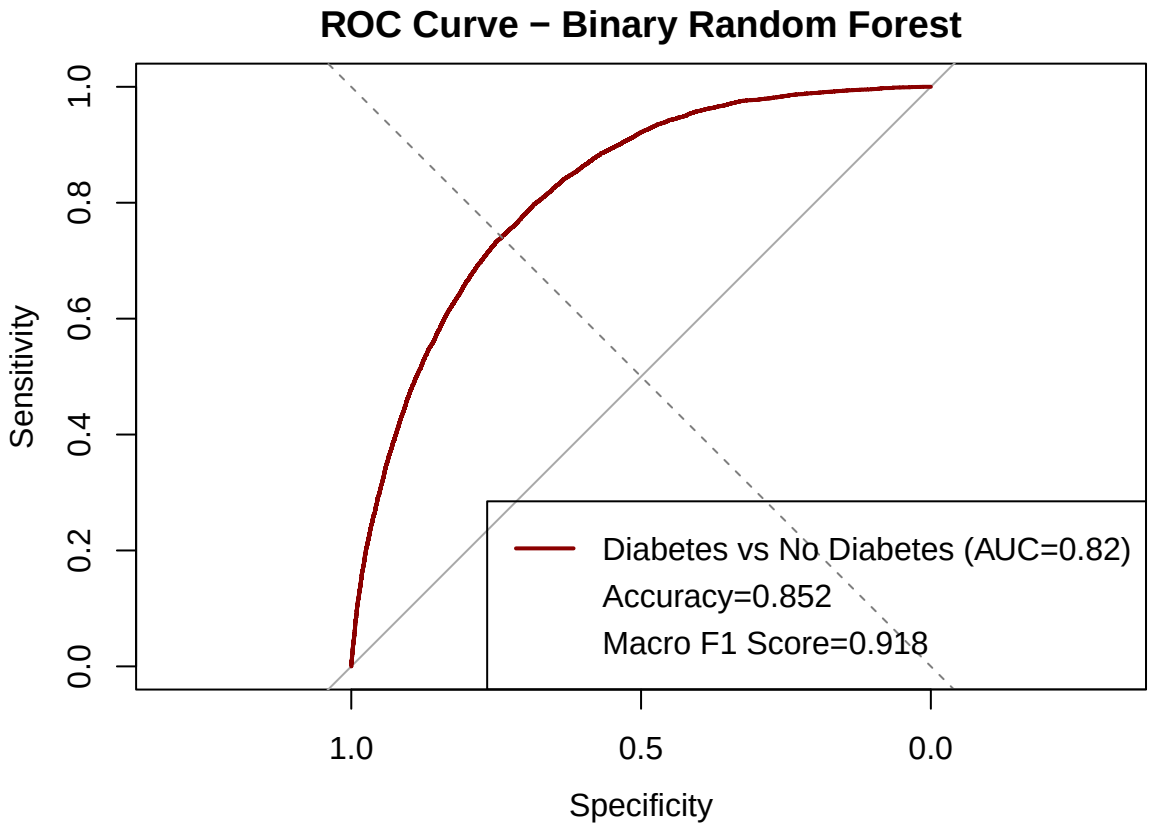
2.4 Confusion matrix heatmap

```r
model_cm_table <- cm$table

model_cm_df <- as.data.frame(model_cm_table) |>
  rename(True = Reference, Predicted = Prediction, Count = Freq)

p_heatmap <- ggplot(model_cm_df, aes(x = Predicted, y = True, fill = Count)) +
  geom_tile(color = "white") +
  geom_text(aes(label = Count), color = "black", size = 5) +
  scale_fill_gradient(low = "#e0f2f1", high = "#00796b") +
  labs(
    title = "Confusion Matrix Heatmap - Random Forest",
    x = "Predicted Class",
    y = "True Class"
  ) +
  theme_minimal(base_size = 14) +
  theme(plot.title = element_text(hjust = 0.5))

print(p_heatmap)
```



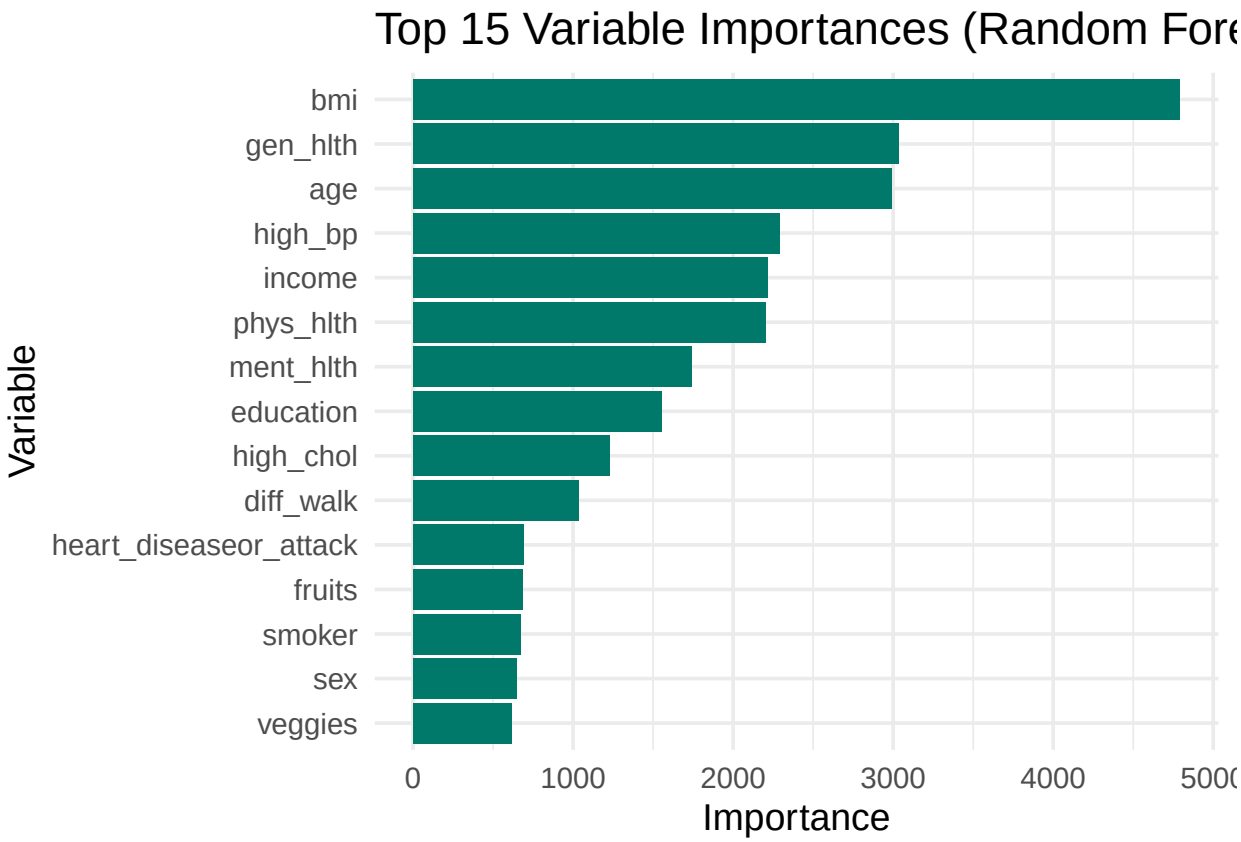
2.5 ROC curve

```
plot(model_roc,
     col = "darkred",
     main = "ROC Curve - Binary Random Forest",
     lwd = 2)

abline(a = 0, b = 1, lty = 2, col = "grey50")

legend(
  "bottomright",
  legend = c(
    paste0("Diabetes vs No Diabetes (AUC=", round(model_auc, 3), ")"),
    paste0("Accuracy=", round(model_accuracy, 3)),
    paste0("Macro F1 Score=", round(model_macro_f1, 3))
  ),
  col = c("darkred", rep(NA, 2)),
  lwd = c(2, rep(NA, 2))
)
```



2.6 Variable importance

```
importance_df <- data.frame(
  Variable = names(rf_model$variable.importance),
  Importance = rf_model$variable.importance
) |>
  arrange(desc(Importance))

ggplot(importance_df[1:15, ], aes(x = reorder(Variable, Importance), y = Importance)) +
  geom_col(fill = "#00796b") +
  coord_flip() +
  labs(title = "Top 15 Variable Importances (Random Forest)",
       x = "Variable",
       y = "Importance") +
  theme_minimal(base_size = 14)
```



Model 3: Binary Logistic Regression Model

```
diabetes_temp = diabetes %>%
  mutate(
    diabetes_binary = ifelse(diabetes_012 == "No Diabetes", 0, 1),
    diabetes_binary = as.numeric(diabetes_binary)
  )
```

We fit a logistic regression model using health and lifestyle predictors such as blood pressure, cholesterol, BMI, smoking status, physical activity, age, and sex.

3.1 Fit the Model

```
logit_model = glm(
  diabetes_binary ~ high_bp + high_chol + bmi + smoker + phys_activity + age + sex,
  data = diabetes_temp,
  family = binomial(link = "logit")
)

summary(logit_model)
```

```
##
## Call:
## glm(formula = diabetes_binary ~ high_bp + high_chol + bmi + smoker +
##     phys_activity + age + sex, family = binomial(link = "logit"),
##     data = diabetes_temp)
##
```

```
## Coefficients:
##                     Estimate Std. Error  z value Pr(>|z|)
## (Intercept)      -4.9335132  0.0337506 -146.175  < 2e-16 ***
## high_bpYes        0.9334468  0.0133878   69.724  < 2e-16 ***
## high_cholYes      0.6788247  0.0125063   54.279  < 2e-16 ***
## bmi               0.0723489  0.0008718   82.984  < 2e-16 ***
## smokerYes         0.1132249  0.0119856    9.447  < 2e-16 ***
## phys_activityYes -0.3568117  0.0128025  -27.870  < 2e-16 ***
## age.L             2.4501416  0.0602296   40.680  < 2e-16 ***
## age.Q            -0.5788621  0.0574324  -10.079  < 2e-16 ***
## age.C            -0.1757155  0.0538836   -3.261  0.00111 **
## age^4             0.0511402  0.0511210    1.000  0.31713
## age^5            -0.1167826  0.0483849   -2.414  0.01580 *
## age^6             0.0237069  0.0449595    0.527  0.59799
## age^7             0.0624394  0.0414595    1.506  0.13206
## age^8            -0.0433950  0.0375788   -1.155  0.24818
## age^9             0.0397879  0.0332257    1.198  0.23111
## age^10           -0.0698033  0.0289679   -2.410  0.01597 *
## age^11            0.0642426  0.0254442    2.525  0.01158 *
## age^12            0.0188388  0.0217445    0.866  0.38629
## sexMale           0.1436624  0.0119704   12.001  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 221031  on 253679  degrees of freedom
## Residual deviance: 185540  on 253661  degrees of freedom
## AIC: 185578
##
## Number of Fisher Scoring iterations: 6
```

3.2 Odds Ratios and 95% Confidence Intervals

```
exp(cbind(
  OR = coef(logit_model),
  confint.default(logit_model)
))
```

```
##                            OR        2.5 %       97.5 %
## (Intercept)       0.007201159  0.006740217  0.007693625
## high_bpYes        2.543260320  2.477394210  2.610877603
## high_cholYes      1.971559144  1.923819868  2.020483062
## bmi               1.075030367  1.073194950  1.076868923
## smokerYes         1.119883711  1.093882640  1.146502816
## phys_activityYes  0.699904251  0.682560422  0.717688787
## age.L            11.589987419 10.299483382 13.042188952
## age.Q             0.560535820  0.500860543  0.627321138
## age.C             0.838856596  0.754782668  0.932295373
## age^4             1.052470431  0.952128709  1.163386837
## age^5             0.889778609  0.809276024  0.978289173
## age^6             1.023990135  0.937618435  1.118318237
## age^7             1.064429908  0.981356280  1.154535872
## age^8             0.957533083  0.889542471  1.030720438
```

```
## age^9             1.040590090  0.974984879  1.110609773
## age^10            0.932577285  0.881104300  0.987057255
## age^11            1.066351021  1.014476588  1.120878012
## age^12            1.019017415  0.976500927  1.063385056
## sexMale           1.154494301  1.127723237  1.181900884
```

3.3 Predicted Probability, Class Prediction, and Confusion Matrix

```
# Predicted probability
diabetes_temp$pred_prob <- predict(logit_model, type = "response")

# Convert to class prediction at threshold = 0.5
diabetes_temp$pred_class <- ifelse(diabetes_temp$pred_prob > 0.5, 1, 0)

# Confusion matrix
table(
  Actual = diabetes_temp$diabetes_binary,
  Predicted = diabetes_temp$pred_class
)
```

```
##       Predicted
## Actual      0      1
##      0 210761   2942
##      1  36471   3506
```

3.4 ROC Curve and AUC

```
library(pROC)

roc_obj <- roc(diabetes_temp$diabetes_binary, diabetes_temp$pred_prob)
```

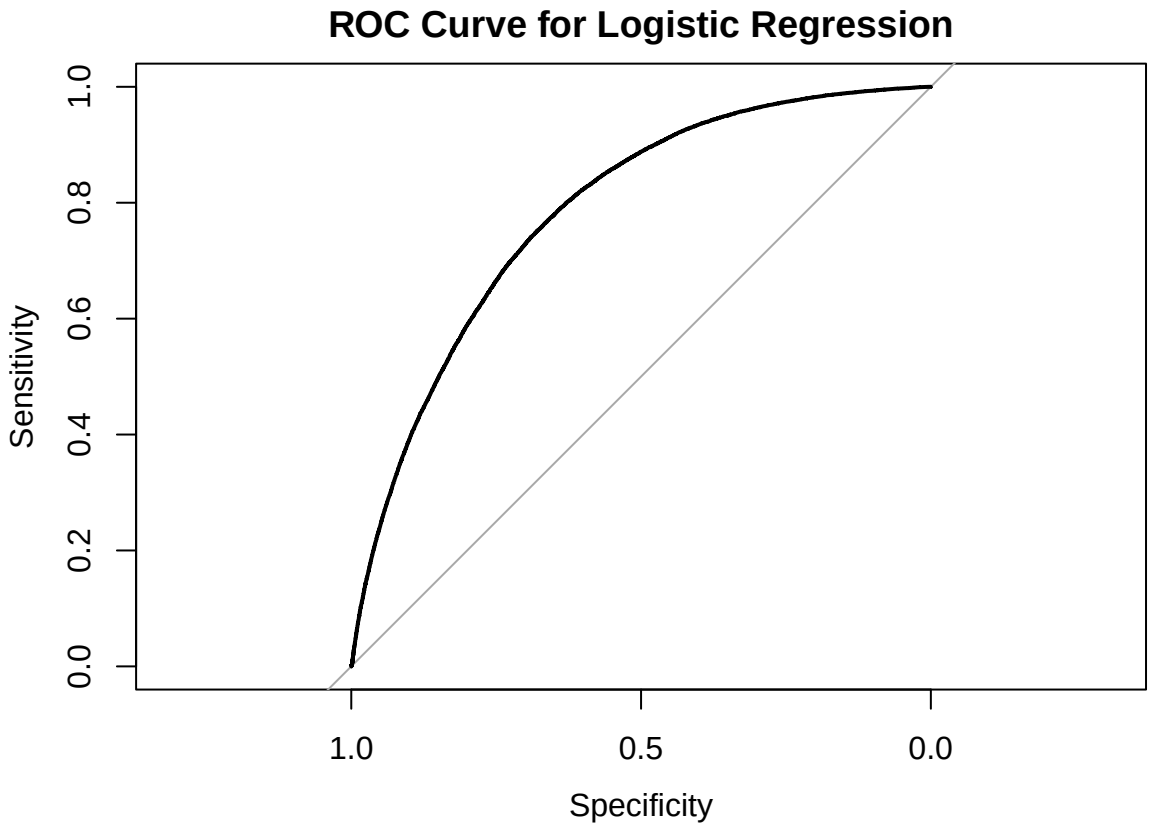
```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```
auc(roc_obj)
```

```
## Area under the curve: 0.7845
```

```
plot(roc_obj, main = "ROC Curve for Logistic Regression")
```



Model 4: Binary XGBoost

We trained an XGBoost model using a 70/30 train-test split. Predictors were converted to numeric matrices using model.matrix.

4.1 Train/Test Split and Data Preparation

```
library(xgboost)
library(caret)
library(dplyr)

#Train/Test Split
set.seed(123)

train_index = createDataPartition(diabetes_temp$diabetes_binary, p = 0.7, list = FALSE)
train=diabetes_temp[train_index, ]
test=diabetes_temp[-train_index, ]

train$binary_outcome=as.numeric(train$diabetes_binary)
test$binary_outcome=as.numeric(test$diabetes_binary)

x_train <- model.matrix(diabetes_binary ~ . - 1, data = train)
x_test  <- model.matrix(diabetes_binary ~ . - 1, data = test)

dtrain=xgb.DMatrix(data = as.matrix(x_train), label = train$diabetes_binary)
dtest=xgb.DMatrix(data = as.matrix(x_test),  label = test$diabetes_binary)
```

4.2 Train the XGBoost Model

```
# Set Hyperparameters

params = list(
  objective = "binary:logistic",
  eval_metric = "auc",
  max_depth = 4,
  eta = 0.1,
  subsample = 0.8,
  colsample_bytree = 0.8
)
```

```
# Train Model

xgb_model = xgb.train(
  params = params,
  data = dtrain,
  nrounds = 200,
  watchlist = list(train = dtrain, test = dtest),
  verbose = 0
)
```

4.3 Predictions and Confusion Matrix

```
pred_prob <- predict(xgb_model, dtest)
pred_class <- ifelse(pred_prob >= 0.5, 1, 0)

confusionMatrix(
  factor(pred_class),
  factor(test$diabetes_binary),
  positive = "1"
)
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction     0     1
##          0 64016     0
##          1     0 12088
##
##                Accuracy : 1
##                  95% CI : (1, 1)
##     No Information Rate : 0.8412
##     P-Value [Acc > NIR] : < 2.2e-16
##
##                   Kappa : 1
##
##  Mcnemar's Test P-Value : NA
##
##             Sensitivity : 1.0000
##             Specificity : 1.0000
##          Pos Pred Value : 1.0000
##          Neg Pred Value : 1.0000
```

```
##              Prevalence : 0.1588
##          Detection Rate : 0.1588
##    Detection Prevalence : 0.1588
##       Balanced Accuracy : 1.0000
##
##        'Positive' Class : 1
##
```

4.4 AUC

```
xgb_auc = roc(test$diabetes_binary, pred_prob)
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```
auc(xgb_auc)
```

```
## Area under the curve: 1
```

4.5 Variable Importance Plot

```
importance_matrix <- xgb.importance(model = xgb_model)
xgb.plot.importance(importance_matrix)
```